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Bridge performance prediction based on nonlinear stochastic processes and digital twins

Hongshuo Sun [a], Li Song [a,b,*], Chenxing Cui [b,c], Zhiwu Yu [a,b]

[a] School of Civil Engineering, Central South University, Changsha 410075, China
[b] National Engineering Research Center of High-speed Railway Construction Technology, Changsha 410075, China
[c] School of Civil Engineering, Henan University of Technology, Zhengzhou 450001, China

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ABSTRACT

Bridge performance prediction helps to develop preventive maintenance measures for bridges. To fully utilize the bridge monitoring data and consider the uncertainty in the bridge performance degradation process, as well as to overcome the limitations of using the Gamma process to predict bridge performance, this paper proposes a bridge performance prediction method based on nonlinear stochastic processes. Considering that the digital twin technology can obtain the degradation data of key performance indexes such as stress levels, stiffness levels, and crack severity of bridges through monitoring data, this paper adopted a generalized nonlinear stochastic process to model the degradation data of bridge performance indexes obtained by the digital twin technology, defined the remaining time (RT) for the bridge performance index to reach the failure threshold, and then obtained the probability density function of the RT associated with the degradation model parameters. In the estimation of the parameter values of the bridge performance degradation model, this paper constructed the degradation model as a state-space model, updated the key parameters of the drift function in the state-space model by the Kalman filtering algorithm, and used the expectation maximization (EM) algorithm combined with the Rauch-Tung-Striebel (RTS) smoothing algorithm to update the hidden states and other parameters in the state space model. Finally, this paper predicted the RT and the RT probability density function for the bridge performance index to reach the failure threshold and the value of the bridge performance index at the next sampling point based on the estimated values of the degradation model parameters. The feasibility and superiority of the method of this paper were verified by numerical simulation and structural tests, respectively.

1. Introduction

Bridges are an important part of transportation infrastructure and are closely related to social and economic development. When the performance of a bridge fails to meet the service requirements, it may affect the efficiency of traffic operation and even impact traffic safety, leading to major casualties. As bridges are subjected to a combination of vehicle loads and the environment, their performance tends to deteriorate with increased service time [1]. Therefore, it is necessary to evaluate and predict the performance of bridges to develop rational bridge operation and maintenance measures. The prediction of bridge performance helps to develop preventive maintenance measures for bridges [2], which is important for improving the operational safety of bridges, prolonging the service life of bridges, and reducing the cost of bridge maintenance.

For the performance assessment of bridges, the assessment includes key performances such as stress levels, stiffness levels, and crack severity of the current bridge [3,4]. When the performance index of a bridge reaches a set limit, the performance index of the bridge is considered to be unable to meet the service requirements. To evaluate the stress level and stiffness level of the current bridge, model updating techniques are usually used to establish a finite element model (FEM) that is consistent with the mechanical behavior of the current bridge [5–9], and the current bridge FEM is used to obtain the stress value and deflection value of the bridge under the specified load to realize the evaluation of stress levels and stiffness levels [10]. To assess the crack severity of the current bridge, scholars have proposed a series of high-precision crack detection methods based on image processing algorithms, which can distinguish the cracks from the background in the image to realize the assessment of the crack severity of bridges [11–14]. To assess the bridge crack severity more comprehensively, some scholars apply digital twin

* Corresponding author at: School of Civil Engineering, Central South University, Changsha 410075, China.
E-mail address: songli@csu.edu.cn (L. Song).

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technology to the assessment of bridge crack severity [15,16]. For example, Shim et al. [17] detected cracks in 3D scanned bridge crack images by image processing algorithms and realized the comprehensive assessment of bridge crack severity by mapping the detected crack geometric information to the corresponding locations in the bridge building information model (BIM). Both FEM and BIM, which are widely used in bridge operations and maintenance, can be considered as digital twin models [18]. It can be seen that the digital twin technology can make full use of the monitoring data to obtain the key performance indexes such as stress levels, stiffness levels, and crack severity in different regions of the bridge, which can realize the comprehensive assessment of bridge performance [19,20].

The prediction of bridge performance mainly includes prediction methods based on accelerated tests and prediction methods based on physical models. Prediction methods based on accelerated tests accelerate structural performance degradation through laboratory structural testing and use laboratory structural performance degradation patterns to predict the performance of bridges in service [21−23]. For example, Song et al. [24] carried out fatigue tests on reinforced concrete plate girders under different load amplitudes and used the test data to establish the relationship between the stress amplitude of the reinforcement and the fatigue life (S-N) to achieve the prediction of the performance of reinforced concrete plate girders in service. However, the prediction method based on accelerated tests makes it difficult to accurately simulate the external environment of bridges in service due to the limitation of structural test conditions in the laboratory on the one hand, and on the other hand, the relationship between accelerated degradation and actual degradation has not yet been fully understood, which limits the prediction accuracy of the method. The prediction method based on the physical model is to establish a bridge performance degradation model based on the material performance degradation pattern [25,26], which is highly scientific and logical. For example, in the study of predicting the performance of bridges under cyclic vehicle loading, scholars have established fatigue damage constitutive models for different materials, in which the damage variable is defined as a function of the number of cyclic loading cycles [27,28], and finite element simulation of bridges under cyclic vehicle loading is carried out by these materials fatigue damages constitutive models to realize the prediction of bridge performance [29]. However, these methods can only realize the accurate prediction of bridge performance in the ideal state, and more complex models are needed when considering the effects of vehicle loading time sequence, environmental loading, and vehicle loading randomness on bridge performance. In addition, both methods do not fully utilize the monitoring data that can reflect the real state of the bridge limiting the accuracy of the prediction of bridge performance. To fully utilize the monitoring data and consider the uncertainty in the degradation process of structural performance, scholars have applied the Gamma process [30], which is an independent incremental property of stochastic processes, to the prediction of bridge performance. For example, Strauss et al. [31] used the Gamma process to describe the growth of creep deformations of a prestressed concrete bridge during long-term service, and the prediction of creep deformations of the prestressed concrete bridge was achieved by modeling the data of creep deformations growth of the prestressed concrete bridge through the Gamma process. Xu et al. [32] used the Gamma process to describe the growth of bearing wear thickness of bridges due to long-term dynamic loading, and the prediction of the bearing wear thickness of bridges was achieved by modeling the bridge bearing wear thickness growth data through the Gamma process. However, the method only investigates some of the bridge performance indexes that can be easily monitored directly, and the prediction of key performance indexes such as stress levels and stiffness levels of the bridge is difficult to realize by relying on this method alone. In addition, since the Gamma process is a class of monotonic stochastic processes with non-negative increments, it can only describe the case where the degradation process of bridge performances is strictly monotonic. However, since bridges are subject to stress redistribution, the changes in key performance indexes such as stress levels and crack severity in some regions of the bridge are not monotonic.

Among stochastic processes, Brownian motion (BM) with drift is a stochastic process with Gaussian noise, where the variance of the noise is a function of time [33]. Since its cumulative degradation is infinitely divisible, this is a required property for any physical degradation process. This property indicates that the degradation process modeled by the BM with drift is not limited by the sampling frequency and sampling interval, and is a necessary property for a stochastic process to be able to model a physical degradation process. The only other stochastic process with this property is the Gamma process. BM with drift was originally used to describe the random swimming of tiny particles in fluids and air; the swimming has a tendency and is suitable for describing a dynamic process with an increasing or decreasing tendency. Since the random term of the BM with drift is Gaussian distributed, it can describe a non-monotonically degradation process, and it has been applied in product degradation modeling [34,35] and further extended to the estimation of the remaining useful life of products [36]. BM with linear drift is a special kind of diffusion process, and the existence of an analytic solution of the probability density distribution of the first hit time, which is called the inverse Gaussian distribution, is its significant advantage. Therefore, BM with linear drift is widely used in the estimation of the remaining useful life of products. However, BM with linear drift can only describe the linear drifted diffusion process and cannot accurately model degradation data that contains nonlinear degradation processes. Therefore, scholars have approximated the nonlinear degradation process as a linear degradation process by some kind of transformation, such as logarithmic transformation or transformation of time scales [37,38]. However, these methods are limited to nonlinear degradation processes that are suitable for transformation in these ways, and many nonlinear degradation processes cannot be transformed in these ways. Further, Si et al. [39] used a time-variable drift coefficient to account for the nonlinearity of the degradation process and applied it to the estimation of the remaining useful life of the product [40]. It can be seen that compared with the Gamma process, which is only capable of describing the strictly monotonic case of the degradation process, the BM with drift can describe the non-monotonous case of the degradation process, which is more helpful for realizing the comprehensive prediction of stress levels, stiffness levels, and crack severity and other key performance indexes in different regions of the bridge.

To overcome these limitations, this paper proposes a bridge performance prediction method based on nonlinear stochastic processes. To achieve an accurate description of the bridge performance degradation process, this paper adopts a generalized nonlinear stochastic process to model the bridge performance degradation data. To realize the prediction of bridge performance through the established degradation model, this paper defines the remaining time (RT) for the bridge performance index to reach the failure threshold, and then obtains the probability density function of the RT associated with the parameters of the degradation model. To realize the estimation of the bridge performance degradation model parameters, this paper constructs the degradation model as a state-space model, updates the key parameters of the drift function in the state-space model by the Kalman filtering algorithm, and adopts the expectation-maximization algorithm combined with the Rauch-Tung-Striebel smoothing algorithm to update the hidden state and other parameters in the state-space model. Considering that the digital twin technology can obtain the degradation data of key performance indexes such as stress levels, stiffness levels, and crack severity of bridges through monitoring data, this paper adopted a generalized nonlinear stochastic process to model the degradation data of bridge performance indexes obtained by the digital twin technology. Based on the estimated values of the degradation model parameters, the method of this paper can predict the RT and the RT probability density function for the bridge performance indexes to reach the failure threshold, as well as the bridge performance index value at the next sampling point. In this

paper, numerical simulations and structural tests are carried out to verify the feasibility and superiority of the method.

The advantages of the method of this paper are as follows: (1) The nonlinear stochastic process used in the method of this paper can describe non-monotonic degradation processes, overcoming the limitation of the Gamma process, which can only describe strictly monotonic degradation processes, and thus has a wider range of applications; (2) The method of this paper adopts a general nonlinear stochastic process with time-varying drift coefficients to model the degradation data of bridge performance indexes, which can not only consider the uncertainty in the degradation process of bridge performance but also accurately track nonlinear degradation trajectories, overcoming the limitation of the BM with linear drift that is difficult to accurately track nonlinear degradation trajectories; (3) The method of this paper adopts digital twin technology to obtain degradation data for bridge performance indexes, which can not only fully utilize monitoring data that reflect the actual service state of bridges but also avoid the high-cost processes of obtaining bridge performance degradation data, such as arranging a large number of sensors and conducting on-site load tests.

2. Methodology

2.1. Bridge performance degradation model based on nonlinear stochastic processes

Assume that $\{X(t),\ t \geq 0\}$ represents a bridge performance degradation process that varies with service time t, and the initial state $X(0)$ is taken to be x_0. When a linear stochastic process with a constant drift coefficient is used to model the bridge performance degradation data, the values of the established bridge performance degradation model at time t can be expressed as follows

$$X(t) = x_0 + \lambda t + \sigma_B B(t) \tag{1}$$

where λ and σ_B denote the drift and diffusion coefficients, respectively, which are constants; and $B(t)$ denotes standard BM. It can be seen that it is difficult to accurately track the bridge performance containing nonlinear degradation processes through the linear stochastic process-based degradation model of Eq. (1). When a nonlinear stochastic process with time-varying drift coefficients is used to model the bridge performance degradation data, the drift coefficient λ in Eq. (1) is replaced by $\lambda{\cdot}\mu(t;\vartheta)$, and $\mu(t;\vartheta)$ is a function of time t and $\mu(t;\vartheta)$ contains the unknown parameter ϑ. The values of the established bridge performance degradation model at time t can be expressed as follows

$$X(t) = x_0 + \lambda \cdot \int_0^t \mu(\tau;\vartheta)d\tau + \sigma_B B(t) \tag{2}$$

where $\mu(t;\vartheta)$ can be used to characterize the nonlinear degradation of bridge performance, λ is a scaling parameter that controls the rate of nonlinear degradation, and ϑ can be used to determine the shape of the nonlinear degradation of bridge performance. It can be seen that the nonlinear stochastic process-based degradation model of Eq. (2) contains nonlinear structures, and the Eq. (2) degradation model is equivalent to the Eq. (1) degradation model when $\mu(t;\vartheta)$ is constant. Therefore, the method in this paper uses the nonlinear stochastic process-based bridge performance degradation model of Eq. (2), and it is known that the parameters of the bridge performance degradation model based on the nonlinear stochastic process include $\boldsymbol{\varphi} = [\lambda, \vartheta, \sigma_B]$.

2.2. Bridge performance prediction based on bridge performance degradation models

2.2.1. Definition of the RT

When a performance index of a bridge reaches the failure threshold that cannot meet the service requirements, the bridge needs to be repaired or demolished. Therefore, predicting the RT and RT probability density function for bridge performance indexes to reach the failure threshold is the most important objective of bridge performance prediction. For predicting the RT for the performance index value x_i at time t_i to reach the failure threshold w, the value of the degradation model at time t when using the linear stochastic process-based degradation model of Eq. (1) can be expressed as

$$X(t) = x_i + \lambda(t - t_i) + \sigma_B(B(t) - B(t_i)) \tag{3}$$

where $t \geq t_i$. Since $\{B(t), t \geq 0\}$ in the stochastic process is a standard BM, taking $W(t - t_i) = B(t) - B(ti)$, $\{W(t - t_i), t - t_i \geq 0\}$ is still a standard BM and the degradation model of Eq. (3) can be written as

$$X(t) = x_i + \lambda(t - t_i) + \sigma_B W(t - t_i) \tag{4}$$

The need to assume is that when the bridge performance index happens to reach the failure threshold w at a certain time t, then the probability that the value of the performance index before the time t exceeds the failure threshold w is negligible. To establish the connection between the degradation model $X(t)$ of Eq. (4) and the RT from the bridge performance index value x_i to reach the failure threshold w, two methods can be chosen as follows. One method defines the RT $Li = \{l_i : X(l_i + t_i) \geq w | X(t_i) = xi\}$, then the cumulative density function of the RT can be expressed as $f_{L_i|x_i}(l_i|xi) = \Pr(X(l_i + t_i) \geq w | X(t_i) = xi)$. The explicit form of the probability density function of the RT can be expressed as [41],

$$f_{L_i|x_i}(l_i|xi) = \frac{1}{\sigma_B\sqrt{2\pi l_i}} \times \exp\left[-\frac{(w - x_i - \lambda l_i)^2}{2\sigma_B^2 l_i}\right] \tag{5}$$

Another method defines the RT $Li = \inf\{l_i : X(l_i + t_i) \geq w | X(t_i) = xi\}$, which is based on the concept of first hit time. It can be understood that, in the degradation model of Eq. (4), when the performance index value $X(t)$ at time t reaches the failure threshold w, it is considered that the performance index fails to satisfy the service requirements of the bridge from time t onwards. The explicit form of the probability density function of the RT can be expressed as [42],

$$f_{L_i|x_i}(l_i|xi) = \frac{w - x_i}{\sigma_B\sqrt{2\pi l_i^3}} \exp\left[-\frac{(w - x_i - \lambda l_i)^2}{2\sigma_B^2 l_i}\right] \tag{6}$$

It can be seen through Eqs. (5) and (6) that the above two methods correspond to different bridge performance prediction results. In contrast to the Eq. (6) method, the Eq. (5) method ignores the probability that the bridge performance index reaches the failure threshold in the interval from time t_i to time t. From the consideration of service safety of bridges, the method in this paper is based on the concept of first hit time to define the RT for bridge performance indexes to reach the failure threshold.

2.2.2. Approximate calculation of the RT probability density function

When establishing the connection between the degradation model $X(t)$ of Eq. (2) and the RT from the initial time to reach the failure threshold w, the first hit time-based concept defines the RT $T = \inf\{t : X(t) \geq w | X(0) < w\}$. When $\mu(t;\vartheta)$ in the degradation model $X(t)$ of Eq. (2) is a continuous function of time t, the explicit form of the probability density function of the RT can be approximated as [39],

$$f_{T|\phi}(t|\phi) \cong \frac{1}{\sqrt{2\pi t}}\left(\frac{S_B(t)}{t} + \frac{\lambda}{\sigma_B}\mu(t;\vartheta)\right)\exp\left[-\frac{S_B^2(t)}{2t}\right] \tag{7}$$

where $S_B(t) = \left(w - \lambda \cdot \int_0^t \mu(\tau;\vartheta)d\tau\right)/\sigma_B$. It can be seen that when $\lambda{\cdot}\mu(t;\vartheta)$ is a constant, Eq. (7) is equivalent to Eq. (6) when $t_i = 0$.

For predicting the RT for the performance index value x_i at time t_i to reach the failure threshold w, the value of the degradation model at time t when using the nonlinear stochastic process-based degradation model of Eq. (2) can be expressed as

$$X(t) = x_i + \lambda\left(\int_0^t \mu(\tau;\vartheta)d\tau - \int_0^{t_i} \mu(\tau;\vartheta)d\tau\right) + \sigma_B(B(t) - B(t_i)) \tag{8}$$

Eq. (8) can be further written as

$$X(t) = x_i + \lambda\int_{t_i}^t \mu(\tau;\vartheta)d\tau + \sigma_B W(t - t_i) \tag{9}$$

The explicit form of the probability density function of the RT based on Eq. (7) and Eq. (9) can be written as

$$f_{L_i|x_i,\phi}(l_i|xi,\phi) \cong \frac{w - x_i - \lambda(v(l_i) - li\mu(l_i + t_i;\vartheta))}{\sigma_B\sqrt{2\pi l_i^3}}\exp\left[-\frac{(w - x_i - \lambda v(l_i))^2}{2\sigma_B^2 l_i}\right] \tag{10}$$

where $v(l_i) = \int_{t_i}^{l_i+t_i}\mu(\tau;\vartheta)d\tau$. It can be seen that under the condition that the performance index value x_i at time t_i is the same, the probability density function of the RT calculated with different values of the degradation model parameters $\boldsymbol{\varphi} = [\lambda, \vartheta, \sigma_B]$ will be different, and its physical significance is shown in Fig. 1. As can be seen from Fig. 1, the RT to reach the failure threshold is significantly smaller in Fig. 1(a) than in Fig. 1(b), as inferred from the degradation data from the initial time to the time t_i. It is obvious that when calculating the probability density function of the RT, it is necessary to consider not only the performance index value x_i at the time t_i, but also the degradation data before the time t_i. If the degradation data before the time t_i is not considered, i.e., the degradation model parameters $\boldsymbol{\varphi} = [\lambda, \vartheta, \sigma_B]$ are fixed, the probability density function of the RT calculated in Fig. 1(a) and Eq. 1(b) through Eq. (10) is the same, which is unreasonable. The degradation model parameters $\boldsymbol{\varphi} = [\lambda, \vartheta, \sigma_B]$ in the method of this paper are estimated from the bridge performance degradation data $\mathbf{X}_{0:i} = \{x_0, x_1, \cdots, x_i\}$, and then a reliable probability density function $f_{L_i|\mathbf{X}_{0:i}}(l_i|\mathbf{X}_{0:i})$ of the RT is calculated.

2.3. Estimation of bridge performance degradation model parameters

To accurately estimate the degradation model parameters $\boldsymbol{\varphi} = [\lambda, \vartheta, \sigma_B]$ from the bridge performance degradation data $\mathbf{X}_{0:i} = \{x_0, x_1, \cdots, x_i\}$, the degradation model of Eq. (2) needs to be transformed into a state model [43]. The parameter λ in the drift part $\lambda{\cdot}\mu(t;\vartheta)$ is updated by introducing a random walk model $\lambda_i = \lambda_{i-1} + \eta$, η follows a normal distribution $N(0, Q)$. The degradation model of Eq. (2) can be transformed into the following state model

$$\begin{cases} \lambda_i = \lambda_{i-1} + \eta \\ x_i = x_{i-1} + \lambda_{i-1}\Omega_i(\vartheta) + \sigma_B\varepsilon_i \end{cases} \tag{11}$$

where $\Omega_i(\vartheta) = h(t_i;\vartheta) - h(t_{i-1};\vartheta)$, $h(t_i;\vartheta) = \int_0^{t_i}\mu(\tau;\vartheta)d\tau$, $\varepsilon_i = [B(t_i) - B(t_{i-1})]$, ε_i follows a normal distribution $N(0, t_i - t_{i-1})$. The property of BM requires the use of $t_i - t_{i-1}$ as the variance of ε_i. The initial drift λ_0 is assumed to follow a normal distribution $N(\mu_0, P_0)$. Since λ_i in Eq. (11) follows a Gaussian distribution, this paper makes $\hat{\lambda}_i = \mathrm{E}(\lambda_i|\mathbf{X}_{0:i})$ and $P_{i|i} = \mathrm{var}(\lambda_i|\mathbf{X}_{0:i})$, and $\hat{\lambda}_i$ and $P_{i|i}$ denote the mean and variance of λ_i, respectively. In the framework of Bayesian filtering, $\hat{\lambda}_i$ and $P_{i|i}$ can be estimated by the Kalman filtering algorithm based on bridge performance degradation data $\mathbf{X}_{0:i} = \{x_0, x_1, \cdots, x_i\}$. The Kalman filtering algorithm for estimating $\hat{\lambda}_i$ and $P_{i|i}$ is shown in Fig. 2.

To calculate the probability density function of the RT from the estimated $\hat{\lambda}_i$ and $P_{i|i}$, Eq. (10) can be written as

$$f_{L_i|\mathbf{X}_{0:i}}(l_i|\mathbf{X}_{0:i}) \cong \frac{(w - x_i)\Lambda(l_i) - \alpha i(l_i;\vartheta)\Delta(l_i)}{\sqrt{2\pi l_i^2\Lambda^3(l_i)}}\exp\left[-\frac{(w - x_i - \hat{\lambda}_i v(l_i))^2}{2\Lambda(l_i)}\right] \tag{12}$$

where $\alpha i(l_i;\vartheta) = v(l_i) - l_i\mu(l_i + t_i;\vartheta)$, $v(l_i) = \int_{t_i}^{l_i+t_i}\mu(\tau;\vartheta)d\tau$, $\Lambda(l_i) = P_{i|i}v(l_i)^2 + \sigma_B^2 l_i$, $\Delta(l_i) = P_{i|i}v(l_i)(w - x_i) + \hat{\lambda}_i\sigma_B^2 l_i$. It can be seen that calculating the probability density function $f_{L_i|x_i,\phi}(l_i|\mathbf{X}_{0:i})$ of the RT by Eq. (12) requires the estimation of several unknown parameters, which include σ_B^2, ϑ, μ_0, P_0, and Q.

Let $\boldsymbol{\theta} = [\sigma_B^2, \vartheta, \mu_0, P_0, Q]$, $\boldsymbol{\theta}$ is a vector of unknown parameters. Let $\boldsymbol{\gamma}_i = \{\lambda_0, \lambda_1, \cdots, \lambda_i\}$, $\boldsymbol{\gamma}_i$ is a sequence of scaling parameters from the initial time to the time t_i. To estimate the scaling parameters λ and $\boldsymbol{\theta}$, it is necessary

Algorithm 1: Kalman filtering algorithm

1. Assume μ_0, P_0.

2. State estimation at time t_i

$$\hat{\lambda}_{i|i-1} = \hat{\lambda}_{i-1}$$

$$P_{i|i-1} = P_{i-1|i-1} + Q$$

$$K_i = P_{i|i-1}\Omega_i(\vartheta)\left(\Omega_i(\vartheta)^2 P_{i|i-1} + \sigma_B^2(t_i - t_{i-1})\right)^{-1}$$

$$\lambda_i = \lambda_{i-1} + K_i\left(x_i - x_{i-1} - \hat{\lambda}_{i-1}\Omega_i(\vartheta)\right).$$

3. Updating variance $P_{i|i} = P_{i|i-1} - \Omega_i(\vartheta)K_i P_{i|i-1}$.

Fig. 2. Kalman filtering algorithm.

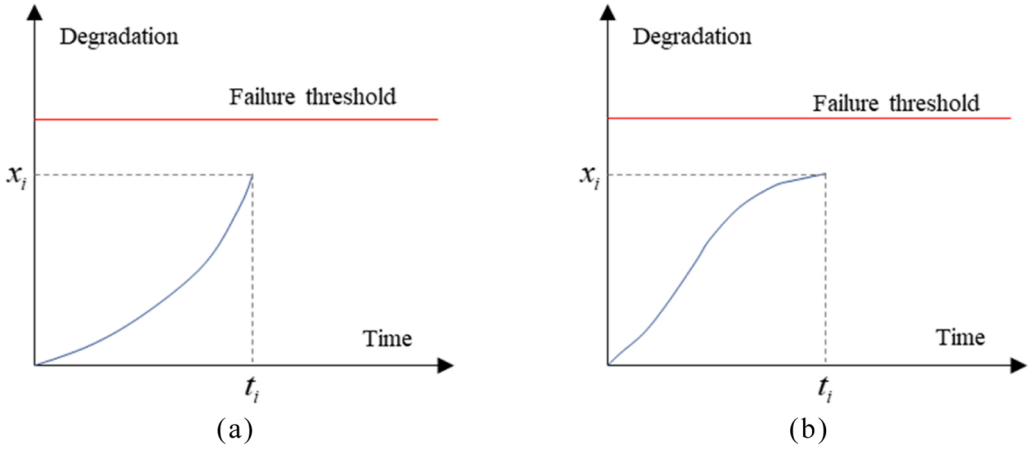


Fig. 1. Different degradation processes with the same amount of degradation.

to be realized by calculating the maximum likelihood estimate of $\boldsymbol{\theta}$. The log-likelihood function of $\boldsymbol{\theta}$ under conditions based on degradation data $\mathbf{X}_{0:i}$ can be expressed as

$$\iota(\boldsymbol{\theta}) = \log[p(\mathbf{X}_{0:i}|\boldsymbol{\theta})] = \sum_{j=1}^{i} \log\left[p\left(x_j|\mathbf{X}_{0:j-1}, \boldsymbol{\theta}\right)\right] \tag{13}$$

Let $\widehat{\boldsymbol{\theta}}_i = \left[\sigma_{B,i}^2, \vartheta_i, \mu_{0,i}, P_{0,i}, Q_i\right]$, $\widehat{\boldsymbol{\theta}}_i$ denotes the maximum likelihood estimate of the t_i moment $\boldsymbol{\theta}$ conditional on the degradation data $\mathbf{X}_{0:i}$. In the state model (11), the data $\boldsymbol{\gamma}_i$ about the parameter λ is used as the state in the state model (11), and it is necessary to obtain $p(\mathbf{X}_{0:i}|\boldsymbol{\theta})$ by missing the data $\boldsymbol{\gamma}_i$. It is difficult to give an explicit form of the function $p(\mathbf{X}_{0:i}|\boldsymbol{\theta})$ due to the presence of hidden variables. The expectation maximization (EM) algorithm [44] provides a reasonable approach for dealing with a parameter estimation problem involving hidden variables.

In this paper, the log-likelihood function for complete data is defined as $\iota(\boldsymbol{\theta}) = \log[p(\mathbf{X}_{0:i}, \boldsymbol{\gamma}_i|\boldsymbol{\theta})]$. In the estimation of parameters, the two steps included in the EM algorithm are specified as follows.

Solve for expectation: Calculate

$$\ell\left(\boldsymbol{\theta}|\widehat{\boldsymbol{\theta}}_i^{(k)}\right) = E_{\boldsymbol{\gamma}_i|\mathbf{X}_{0:i},\widehat{\boldsymbol{\theta}}_i^{(k)}}\{\log p(\mathbf{X}_{0:i}, \boldsymbol{\gamma}_i|\boldsymbol{\theta})\} \tag{14}$$

where $\widehat{\boldsymbol{\theta}}_i^{(k)} = \left[\sigma_{B,i}^{2(k)}, \mu_{0,i}^{(k)}, P_{0,i}^{(k)}, Q_i^{(k)}\right]$ denotes the parameters estimated at the kth iteration step in the condition based on the degradation data $\mathbf{X}_{0:i}$.

Maximize: Calculate

$$\widehat{\boldsymbol{\theta}}_i^{(k+1)} = \text{argmax}_{\boldsymbol{\theta}}\ell\left(\boldsymbol{\theta}|\widehat{\boldsymbol{\theta}}_i^{(k)}\right) \tag{15}$$

Specifically, in the step of solving for the expectation, to obtain the expectation value $\ell(\boldsymbol{\theta})$ of the unknown parameter vector $\widehat{\boldsymbol{\theta}}_i^{(k)}$ at the kth iteration step under the conditions based on the degradation data $\mathbf{X}_{0:i}$, there is

$$\begin{aligned}\ell\left(\boldsymbol{\theta}|\widehat{\boldsymbol{\theta}}_i^{(k)}\right) &= E_{\boldsymbol{\gamma}_i|\mathbf{X}_{0:i},\widehat{\boldsymbol{\theta}}_i^{(k)}}\{\log p(\mathbf{X}_{0:i}, \boldsymbol{\gamma}_i|\boldsymbol{\theta})\} \\ &= E_{\boldsymbol{\gamma}_i|\mathbf{X}_{0:i},\widehat{\boldsymbol{\theta}}_i^{(k)}}\{\log p(\mathbf{X}_{0:i}|\boldsymbol{\gamma}_i, \boldsymbol{\theta})\} \\ &\quad + E_{\boldsymbol{\gamma}_i|\mathbf{X}_{0:i},\widehat{\boldsymbol{\theta}}_i^{(k)}}\{\log p(\boldsymbol{\gamma}_i|\boldsymbol{\theta})\}\end{aligned} \tag{16}$$

According to the Markov property of the standard BM, the complete log-likelihood function can be written as

$$\begin{aligned}\ell(\boldsymbol{\theta}) &= \log p(\mathbf{X}_{0:i}|\boldsymbol{\gamma}_i, \boldsymbol{\theta}) + \log p(\boldsymbol{\gamma}_i|\boldsymbol{\theta}) \\ &= \log p(\lambda_0|\boldsymbol{\theta}) + \log \prod_{j=1}^{i} p\left(\lambda_j|\lambda_{j-1}, \boldsymbol{\theta}\right) + \log \prod_{j=1}^{i} p\left(x_j|\lambda_{j-1}, \boldsymbol{\theta}\right)\end{aligned} \tag{17}$$

The state model of Eq. (11) shows that $\lambda_j|\lambda_{j-1}$ follows a normal distribution $N(\lambda_{j-1}, Q)$, $x_j|\lambda_{j-1}$ follows a normal distribution $N\left(x_{j-1} + \lambda_{j-1}\int_{t_{j-1}}^{t_j}\mu(\tau;\vartheta)d\tau, \sigma_B^2(t_j - t_{j-1})\right)$, λ_0 follows a normal distribution $N(\mu_0, P_0)$. Combining Eq. (17) and neglecting the constant term, the complete log-likelihood function can be written as

Algorithm 2: RTS smoothing algorithm

1. Iterate forward according to Algorithm 1.

2. Backward iteration

$$S_j = P_{j|j}P_{j+1|j}^{-1}$$

$$\hat{\lambda}_{j|i} = \hat{\lambda}_j + S_j\left(\hat{\lambda}_{j+1|i} - \hat{\lambda}_{j+1|j}\right) = \hat{\lambda}_j + S_j\left(\hat{\lambda}_{j+1|i} - \hat{\lambda}_j\right)$$

$$P_{j|i} = P_{j|j} + S_j^2\left(P_{j+1|i} - P_{j+1|j}\right).$$

3. Initialize

$$M_{i|i} = \left(1 - K_i\int_{t_{j-1}}^{t_j}\mu(\tau;\vartheta)d\tau\right)P_{i-1|i-1}.$$

4. Backward iteration for smoothing covariance

$$M_{j|i} = P_{j|j}S_{j-1} + S_j\left(M_{j+1|i} - P_{j|j}\right)S_{j-1}.$$

Fig. 3. RTS smoothing algorithm.

$$\begin{aligned}2\ell(\boldsymbol{\theta}) = &-\log P_0 - \frac{(\lambda_0 - \mu_0)^2}{P_0} - \sum_{j=1}^{i}\left(\log Q + \frac{(\lambda_j - \lambda_{j-1})^2}{Q}\right) \\ &- \sum_{j=1}^{i}\left(2\log\sigma_B + \frac{\left(x_j - x_{j-1} - \lambda_{j-1}\int_{t_{j-1}}^{t_j}\mu(\tau;\vartheta)d\tau\right)^2}{\sigma_B^2(t_j - t_{j-1})}\right)\end{aligned} \tag{18}$$

Based on Eq. (18), the conditional expectation $\ell\left(\boldsymbol{\theta}|\widehat{\boldsymbol{\theta}}_i^{(k)}\right)$ given in Eq. (16) can be further written as

$$\begin{aligned}2\ell\left(\boldsymbol{\theta}|\widehat{\boldsymbol{\theta}}_i^{(k)}\right) &= E_{\boldsymbol{\gamma}_i|\mathbf{X}_{0:i},\widehat{\boldsymbol{\theta}}_i^{(k)}}\left[2\log\ell(\boldsymbol{\theta})\right] \\ = E_{\boldsymbol{\gamma}_i|\mathbf{X}_{0:i},\widehat{\boldsymbol{\theta}}_i^{(k)}} \times &\left[-\log P_0 - \frac{(\lambda_0 - \mu_0)^2}{P_0} - \sum_{j=1}^{i}\left(\log Q + \frac{(\lambda_j - \lambda_{j-1})^2}{Q}\right)\right. \\ &\left. - \sum_{j=1}^{i}\left(2\log\sigma_B + \frac{\left(x_j - x_{j-1} - \lambda_{j-1}\int_{t_{j-1}}^{t_j}\mu(\tau;\vartheta)d\tau\right)^2}{\sigma_B^2(t_j - t_{j-1})}\right)\right]\end{aligned} \tag{19}$$

Calculation of $E_{\boldsymbol{\gamma}_i|\mathbf{X}_{0:i},\widehat{\boldsymbol{\theta}}_i^{(k)}}(\lambda_j)$, $E_{\boldsymbol{\gamma}_i|\mathbf{X}_{0:i},\widehat{\boldsymbol{\theta}}_i^{(k)}}\left(\lambda_j^2\right)$, and $E_{\boldsymbol{\gamma}_i|\mathbf{X}_{0:i},\widehat{\boldsymbol{\theta}}_i^{(k)}}(\lambda_j\lambda_{j-1})$ is required in solving for expectations, these are conditional expectations based on the degradation data $\mathbf{X}_{0:i}$. The above conditional expectation can be optimally estimated by the Rauch-Tung-Striebel (RTS) smoothing algorithm [45], the RTS smoothing algorithm is shown in Fig. 3. Where the covariance of two neighboring drift coefficient historical data is denoted as $M_{j|i} = \text{cov}(\lambda_j, \lambda_{j-1}|\mathbf{X}_{0:i})$.

According to the RTS smoothing algorithm, the conditional expectations $E_{\boldsymbol{\gamma}_i|\mathbf{X}_{0:i},\widehat{\boldsymbol{\theta}}_i^{(k)}}(\lambda_j)$, $E_{\boldsymbol{\gamma}_i|\mathbf{X}_{0:i}\widehat{\boldsymbol{\theta}}_i^{(k)}}\left(\lambda_j^2\right)$, and $E_{\boldsymbol{\gamma}_i|\mathbf{X}_{0:i},\widehat{\boldsymbol{\theta}}_i^{(k)}}(\lambda_j\lambda_{j-1})$ can be calculated by the following equation

$$E_{\boldsymbol{\gamma}_i|\mathbf{X}_{0:i},\widehat{\boldsymbol{\theta}}_i^{(k)}}(\lambda_j) = \widehat{\lambda}_{j|i} \tag{20}$$

$$E_{\gamma_i|\mathbf{x}_{0:i},\hat{\boldsymbol{\theta}}_i^{(k)}}\left(\lambda_j^2\right)=\widehat{\lambda}_{j|i}^2+P_{j|i} \tag{21}$$

$$E_{\gamma_i|\mathbf{x}_{0:i},\hat{\boldsymbol{\theta}}_i^{(k)}}\left(\lambda_j\lambda_{j-1}\right)=P_{j|i}S_{j-1}+S_j\left(M_{j+1|i}-P_{j|j}\right)S_{j-1}+\widehat{\lambda}_{j|i}\widehat{\lambda}_{j-1|i} \tag{22}$$

Combining Eqs. (19), (20), (21), and (22), $\ell\left(\boldsymbol{\theta}|\widehat{\boldsymbol{\theta}}_i^{(k)}\right)$ can be expressed as

$$\begin{aligned}
&2\ell\left(\boldsymbol{\theta}|\widehat{\boldsymbol{\theta}}_i^{(k)}\right)\\
&=E_{\gamma_i|\mathbf{x}_{0:i},\hat{\boldsymbol{\theta}}_i^{(k)}}\left[2\ell_i(\boldsymbol{\theta})\right]\\
&=E_{\gamma_i|\mathbf{x}_{0:i},\hat{\boldsymbol{\theta}}_i^{(k)}}\times\left[-\log P_0-\frac{(\lambda_0-\mu_0)^2}{P_0}-\sum_{j=1}^{i}\left(\frac{\left(\lambda_j-\lambda_{j-1}\right)^2}{Q}\right)\right.\\
&\left.-\sum_{j=1}^{i}\left(\log\sigma_B^2+\frac{\left(x_j-x_{j-1}-\lambda_{j-1}\int_{t_{j-1}}^{t_j}\mu(\tau;\vartheta)d\tau\right)^2}{\sigma_B^2\left(t_j-t_{j-1}\right)}\right)\right]-i\log Q\\
&=-\log P_0-\frac{C_{0|i}-2\widehat{\lambda}_{0|i}\mu_0+\mu_0^2}{P_0}-\sum_{j=1}^{i}\left(\log Q+\frac{C_{j|i}-2C_{jj-1|i}+C_{j-1|i}}{Q}\right)\\
&-\sum_{j=1}^{i}\left(\log\sigma_B^2+\frac{\left(x_j-x_{j-1}\right)^2+\left(\int_{t_{j-1}}^{t_j}\mu(\tau;\vartheta)d\tau\right)^2C_{j-1|i}}{\sigma_B^2\left(t_j-t_{j-1}\right)}\right)+\sum_{j=1}^{i}\left(\frac{2\widehat{\lambda}_{j-1|i}\left(x_j-x_{j-1}\right)\cdot\int_{t_{j-1}}^{t_j}\mu(\tau;\vartheta)d\tau}{\sigma_B^2\left(t_j-t_{j-1}\right)}\right)
\end{aligned} \tag{23}$$

After obtaining $\ell\left(\boldsymbol{\theta}|\widehat{\boldsymbol{\theta}}_i^{(k)}\right)$, the unknown parameter $\widehat{\boldsymbol{\theta}}_i^{(k+1)}$ can be estimated by maximizing $\ell\left(\boldsymbol{\theta}|\widehat{\boldsymbol{\theta}}_i^{(k)}\right)$ related to $\boldsymbol{\theta}$. However, since the parameter ϑ is unknown, it is difficult to obtain an explicit form of $\widehat{\boldsymbol{\theta}}_i^{(k+1)}$ by maximizing $\ell\left(\boldsymbol{\theta}|\widehat{\boldsymbol{\theta}}_i^{(k)}\right)$. To solve this problem, the unknown parameters $\widehat{\boldsymbol{\theta}}_i^{(k+1)}$ need to be estimated by a profile log-likelihood function technique, which procedure is as follows

When the parameter ϑ is given, the unknown parameters $\widehat{\boldsymbol{\theta}}_i^{(k+1)}$ are calculated as follows

$$\mu_{0,i}^{(k+1)}=\widehat{\lambda}_{0|i} \tag{24}$$

$$P_{0,i}^{(k+1)}=C_{0|i}-\widehat{\lambda}_{0|i}^2=P_{0|i} \tag{25}$$

$$Q_i^{(k+1)}=\frac{1}{i}\sum_{j=1}^{i}\left(C_{j|i}-2C_{jj-1|i}+C_{j-1|i}\right) \tag{26}$$

$$\begin{aligned}
\sigma_{B,i}^{2(k+1)}(\vartheta)&=\frac{1}{i}\sum_{j=1}^{i}\left(\frac{\left(x_j-x_{j-1}\right)^2+\left(\int_{t_{j-1}}^{t_j}\mu(\tau;\vartheta)d\tau\right)^2C_{j-1|i}}{t_j-t_{j-1}}\right)\\
&-\frac{1}{i}\sum_{j=1}^{i}\left(\frac{2\widehat{\lambda}_{j-1|i}\left(x_j-x_{j-1}\right)\int_{t_{j-1}}^{t_j}\mu(\tau;\vartheta)d\tau}{t_j-t_{j-1}}\right)
\end{aligned} \tag{27}$$

where $C_{j|i}=E_{\gamma_i|\mathbf{x}_{0:i},\hat{\boldsymbol{\theta}}_i^{(k)}}\left(\lambda_j^2\right)$, $C_{jj-1|i}=E_{\gamma_i|\mathbf{x}_{0:i},\hat{\boldsymbol{\theta}}_i^{(k)}}\left(\lambda_j\lambda_{j-1}\right)$. It can be seen that $\sigma_{B,i}^{2(k+1)}(\vartheta)$ is a function of ϑ and it is necessary to obtain $\vartheta_i^{(k)}$ before the unknown parameters $\widehat{\boldsymbol{\theta}}_i^{(k+1)}$ are calculated. To obtain $\vartheta_i^{(k)}$, combining Eqs. (23), (24), (25), (26), and (27), the profile log-likelihood function is obtained as follows

$$\begin{aligned}
&2\ell\left(\boldsymbol{\theta}|\widehat{\boldsymbol{\theta}}_i^{(k)}\right)\\
&=-\log P_{0|i}-1-\sum_{j=1}^{i}\left(\log Q_i^{(k+1)}+\frac{C_{j|i}-2C_{jj-1|i}+C_{j-1|i}}{Q_i^{(k+1)}}\right)\\
&-\sum_{j=1}^{i}\left(\log\sigma_{B,i}^{2(k+1)}(\vartheta)+\frac{\left(x_j-x_{j-1}\right)^2+\left(\int_{t_{j-1}}^{t_j}\mu(\tau;\vartheta)d\tau\right)^2C_{j-1|i}}{\sigma_{B,i}^{2(k+1)}(\vartheta)\left(t_j-t_{j-1}\right)}\right)\\
&+\sum_{j=1}^{i}\left(\frac{2\widehat{\lambda}_{j-1|i}\left(x_j-x_{j-1}\right)\cdot\int_{t_{j-1}}^{t_j}\mu(\tau;\vartheta)d\tau}{\sigma_{B,i}^{2(k+1)}(\vartheta)\left(t_j-t_{j-1}\right)}\right)
\end{aligned} \tag{28}$$

The search algorithm is used to maximize the profile log-likelihood function of Eq. (28), which can achieve the acquisition of $\vartheta_i^{(k)}$. The search algorithm used in this paper is the "fminsearch" function in MATLAB. The calculation of the unknown parameters $\mu_{0,i}^{(k+1)}$, $P_{0,i}^{(k+1)}$, $Q_i^{(k+1)}$, and $\sigma_{B,i}^{2(k+1)}\left(\vartheta_i^{(k)}\right)$ can be realized by taking the obtained $\vartheta_i^{(k)}$ into Eq. (27). A search algorithm is used to maximize the profile log-likelihood function, and the estimation of the bridge performance degradation model parameters is finally achieved by iterating the two steps of solving expectation and maximization in the EM algorithm. The pseudocode for estimating the bridge performance degradation model parameters is shown in Fig. 4.

2.4. Framework for bridge performance prediction based on nonlinear stochastic processes

In the prediction of bridge performance, a generalized nonlinear stochastic process is adopted to model the degradation data of bridge performance indexes obtained by the digital twin technology. Then, based on the estimated values of the degradation model parameters, the prediction of bridge performance is achieved. Finally, the prediction information of bridge performance in different regions is visualized, enabling the formulation of reasonable preventive maintenance measures for the bridge. The framework for bridge performance prediction

Pseudocode for estimating bridge performance degradation model parameters

Step 1: Forward iteration by Kalman filtering algorithm.

Step 2: Backward iteration by RTS smoothing algorithm.

Step 3: Assume ϑ, and obtain $\vartheta_i^{(k)}$ by maximizing Eq. (28) using a search algorithm.

Step 4: Calculate $\boldsymbol{\theta}_i^{(k+1)}=[\mu_{0,i}^{(k+1)},P_{0,i}^{(k+1)},Q_i^{(k+1)},\sigma_{B,i}^{2(k+1)}(\vartheta_i^{(k)})]$ according to Eqs. (24)-(27).

Step 5: Calculate the deviation between $\boldsymbol{\theta}_i^{(k+1)}$ and the assumed value or $\boldsymbol{\theta}_i^{(k)}$.

Step 6: Use $\boldsymbol{\theta}_i^{(k+1)}$ to replace the assumed value or $\boldsymbol{\theta}_i^{(k)}$ in Step 1 and Step 3.

Step 7: When the deviation cannot meet the requirement, set k=k+1 and sequentially complete the calculations from Step 1 to Step 6.

Step 8: When the deviation meets the requirement, output the parameters $\boldsymbol{\theta}_i^{(k+1)}$.

Fig. 4. Pseudocode for estimating bridge performance degradation model parameters.

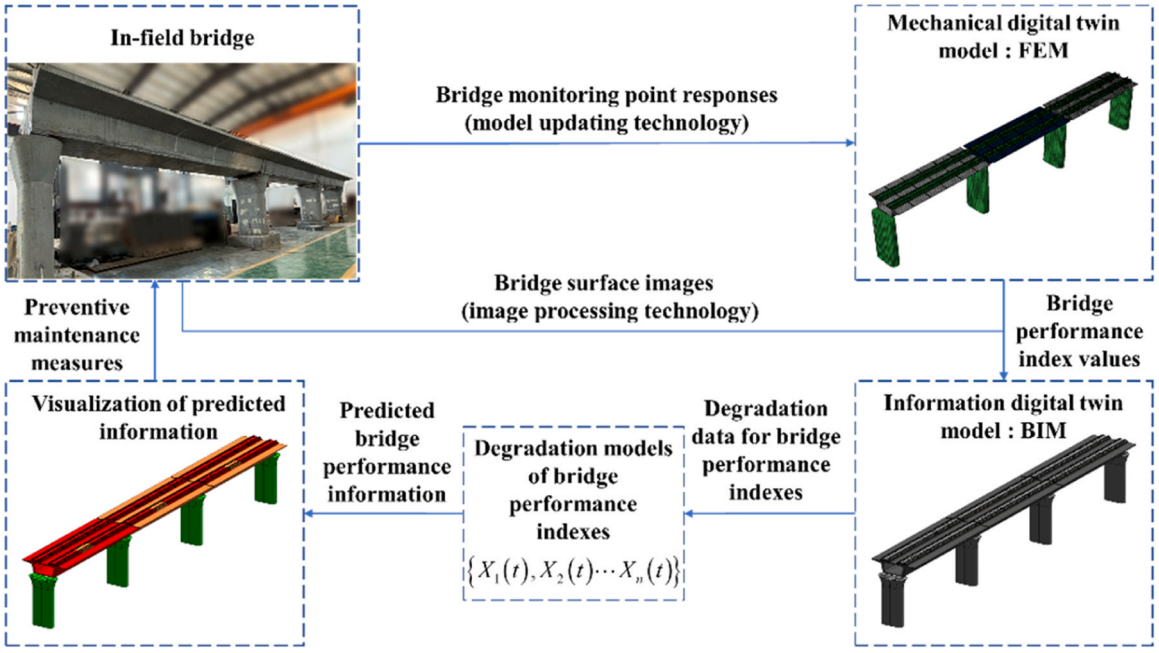


Fig. 5. A framework for bridge performance prediction based on nonlinear stochastic processes.

based on nonlinear stochastic processes is shown in Fig. 5, and the steps are as follows.

Step 1: Build a mechanical digital twin model of the bridge. Obtain the response of the bridge monitoring points, and update the parameters of the baseline FEM of the bridge according to the response of the monitoring points through the model updating technique, so that the updated FEM is consistent with the mechanical behavior of the current bridge. The updated FEM can be used as the current bridge FEM, which can also be regarded as the mechanical digital twin model of the current bridge.

Step 2: Build an information digital twin model of the bridge. The current bridge FEM can be used to obtain stress levels, stiffness levels, and other key performance indexes of the current bridge, and the obtained bridge performance index values of different regions can be mapped to the corresponding locations of the bridge BIM. In addition, the bridge crack detection robot can obtain the bridge surface image, the bridge surface image can be processed based on the image processing technology to obtain the bridge crack information, and the crack information containing the location and geometric size can be mapped to the corresponding location of the bridge BIM. Since the bridge BIM contains the values of performance indexes such as stress levels, stiffness levels, and crack severity of the current bridge, this bridge BIM can be regarded as an information digital twin model of the current bridge.

Step 3: Build degradation models to predict bridge performance. By regularly sampling bridge performance index values to update the bridge information digital twin model, the degradation data of bridge performance indexes can be obtained. After obtaining the degradation data of bridge performance indexes, it is first necessary to determine the specific functional form of the function $\mu(t;\vartheta)$ in Eq. (2) based on prior knowledge, and the specific functional form corresponding to each performance index may be different. Then, the generalized nonlinear stochastic process with the corresponding time-varying function form is used to model the degradation data of the bridge performance index, and the degradation model corresponding to each bridge performance index can be obtained. Finally, the prediction of bridge performance can be realized by the established bridge performance index degradation models $\{X_1(t),X_2(t)\cdots X_n(t)\}$. The predicted bridge performance information includes the RT and the RT probability density function for the

bridge performance index to reach the failure threshold, and the bridge performance index value at the next sampling point. By mapping bridge performance prediction information in different regions to the corresponding locations in the bridge BIM, engineers can visualize the predicted information to formulate reasonable bridge preventive maintenance measures.

For the use of the predicted RT, it can be divided into three levels: safe, medium, and dangerous, based on the predicted RT. Specifically, the larger the predicted RT, the safer the status. When the level is safe, no action is required. When the level is medium, the sampling frequency needs to be increased to improve the accuracy of the prediction results. When the level is dangerous, on-site action is required. For the use of the predicted RT probability density function, the probability of reaching the failure threshold within a specified period can be obtained through integration, it can be divided into three levels: safe, medium, and dangerous, based on the probability value, with a smaller probability value indicating greater safety. Corresponding actions can then be taken in accordance with the above-mentioned rules based on the assigned level. Compared to formulating maintenance measures based solely on the proximity of the current bridge performance index value to the failure threshold, the predicted results of the method of this paper include bridge performance index degradation data from the initial moment to the current moment. This enables the formulation of more reasonable maintenance measures. It is worth noting that these levels are arbitrary and can be properly defined based on the desired limit states of the system of concern.

For traditional methods, the degradation data of some performance indexes of bridges usually requires the installation of a large number of sensors and the conduct of on-site load tests to obtain. For example, the acquisition of bridge strain values and deflection values, which reflect the stress level and stiffness level of bridges, typically requires on-site bridge load tests to obtain the strain values and deflection values of bridges under specified loads [46]. In addition, obtaining strain values of different regions and deflection values of different cross-sections of bridges requires the installation of a large number of sensors, and each sampling requires the completion of an on-site bridge load test, which is costly. The method of this paper uses digital twin technology to obtain degradation data of bridge performance indexes. Model update technology can dynamically update the bridge mechanical twin model based on the bridge monitoring point responses to ensure the mechanical twin model is always consistent with the mechanical behavior of the in-service bridge [18]. The strain values and deflection values of the bridge under specified loads can be easily obtained by simulating on-site load tests through the established bridge mechanical twin model. Furthermore, strain values of different regions and deflection values of different cross-sections of the bridge can also be obtained. In addition, for the bridge strain values and crack geometric dimensions that reflect the stress level and crack severity of bridges, the strain values and crack geometric dimensions corresponding to different regions are different. After completing multiple samplings, the amount of information on the degradation data of bridge performance indexes is very large. The method of this paper is based on the information management advantages of BIM [47]. Through the established bridge information twin model based on BIM, the obtained bridge performance index degradation data information can be effectively stored and managed, and the bridge information digital twin model can be updated once after each sampling. The method of this paper adopts digital twin technology to obtain degradation data for bridge performance indexes, which can not only fully utilize monitoring data that reflect the actual service state of bridges but also avoid the high-cost processes of obtaining bridge performance degradation data, such as arranging a large number of sensors and conducting on-site load tests.

3. Bridge performance prediction study

The bridge performance degradation model of Eq. (2) shows that $\mu(t;\vartheta)$ can be any continuous function of time t, which indicates that the bridge performance degradation model of Eq. (2) has a strong range of applicability. In predicting a performance index of bridges, a specific functional form of $\mu(t;\vartheta)$ needs to be given based on priori information. Scholars typically obtain the performance degradation patterns of in-service bridges by conducting structural fatigue tests. Therefore, this paper conducted fatigue tests on reinforced concrete beams and obtained data on the changes in the mid-span deflection of reinforced concrete beams with increasing number of loading cycles under a constant load, as shown in Section 3.2. Based on these data, it can be seen that the mid-span deflection of reinforced concrete beams increases with the number of cyclic loadings. The growth process of mid-span deflection of reinforced concrete beams can be divided into three stages according to the growth rate. The first stage has a relatively fast growth rate, the second stage has a relatively slow growth rate and the longest duration, and the third stage has an extremely rapid growth rate. Since reinforced concrete beams are quickly damaged in the third stage, for safety reasons, bridges in service should not be allowed to be in the third stage, and the failure threshold of bridge performance indexes should be set in the second stage. Therefore, it is only necessary to analyze the performance degradation pattern of bridges in the first two stages. For the first two stages, the growth pattern of the mid-span deflection of reinforced concrete beams satisfies the power function change pattern with the number of cyclic loadings as the variable, where the exponent in the power function is greater than 0 and less than 1. According to the fatigue data of reinforced concrete beams in References [24] and [48], the degradation patterns of stiffness and the growth patterns of crack severity in these references also follow a power function variation pattern with the number of cyclic loadings as the variable, where the exponent in the power function is greater than 0 and less than 1. Therefore, this paper assumes that the degradation patterns of bridge performance indexes in service follow the power function change pattern with service time as the variable, where the exponent in the power function is greater than 0 and less than 1. Therefore, this paper assumes that the function $\int_0^t \mu(\tau;\vartheta)d\tau = t^a$ in the bridge performance degradation model of Eq. (2). Among them, a should be greater than 0 and less than 1. The corresponding function $\mu(t;\vartheta)$ is defined as $\mu(t;\vartheta) = at^{a-1}$. The bridge performance degradation model of Eq. (2) can be written as

$$X(t) = x_0 + \lambda \cdot t^a + \sigma_B B(t) \tag{29}$$

In this paper, the bridge performance degradation model of Eq. (29) is used in all subsequent studies to verify the feasibility and superiority of this paper's method.

3.1. Numerical study of bridge performance prediction

To validate the feasibility and superiority of the method of this paper in predicting bridge performance indexes containing nonlinear degradation processes, this study uses pure numerical simulation data to substitute for the degradation data of bridge performance indexes containing nonlinear degradation processes. This study generates degradation data containing nonlinear stochastic processes by Euler approximation of the degradation model of Eq. (29) to simulate the degradation data of bridge performance indexes containing nonlinear stochastic processes. The simulated degradation data generation process is as follows

$$X_{(i+1)\Delta t} = X_{i\Delta t} + \lambda a(i\Delta t)^{a-1}\Delta t + \sigma_B Y\sqrt{\Delta t} \tag{30}$$

Table 1
Parameter values.

Parameters	x_0	a	λ	σ_B	Δt
Parameter values	0	0.5	0.5	0.05	0.1

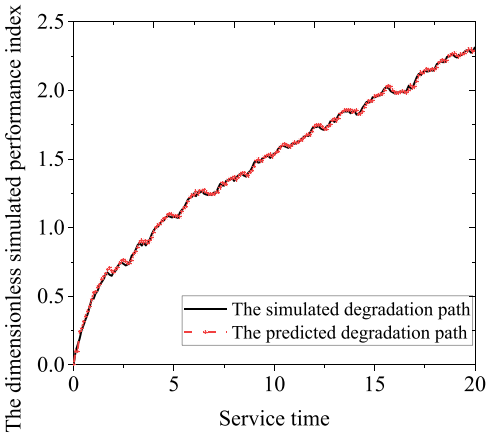


Fig. 6. Simulated degradation paths and predicted degradation paths.

where Y follows a normal distribution $N(0,1)$; Δt is the sampling interval time; i represents the number of samples. When sampling 200 times, the simulated degradation data containing 200 data points can be obtained by Eq. (30). The parameter values for this study are shown in Table 1. By sequentially connecting all data points calculated by Eq. (30), the simulated degradation path can be obtained, as shown in Fig. 6. In Fig. 6, the horizontal coordinate values of the simulated degradation path data points represent the service time t_i, where $t_i = i \times \Delta t$. The vertical coordinate values of the simulated degradation path data points represent the simulated bridge performance index values. Since the simulated degradation path data points are simulated data, neither the horizontal nor vertical coordinate values of the simulated degradation path data points have corresponding physical units. In Eq. (30), $\sigma_B Y\sqrt{\Delta t}$ can be regarded as a random noise term, and Eq. (30) can be further written as

$$X_{(i+1)\Delta t} - X_{i\Delta t} = \left(\lambda a(i)^{a-1}\Delta t^{a-1/2} + \sigma_B Y\right)\sqrt{\Delta t} \tag{31}$$

From Eq. (31), it can be seen that when $a < 1/2$, the level of random noise increases as the time step increases, while when $a > 1/2$, the noise level decreases as the time step increases.

Through the method of this paper, the simulated degradation data is modeled and the parameter values of the degradation model are estimated, and the estimated degradation model parameter values are shown in Fig. 7. From Fig. 7, it can be seen that with the increase of sampling data, the estimated values of a, μ_0, and σ_B are close to the actual values of 0.5, 0.5, and 0.05, respectively, which verifies the feasibility of this paper's method for modeling the degradation data of bridge performance indexes containing nonlinear degradation processes. When modeling simulated degradation data by the method of using linear stochastic processes, it can be seen from Eq. (1) that the degradation model established by the method of using linear stochastic processes does not include parameter a. The parameter values of the degradation model estimated by the method of using linear stochastic processes are shown in Fig. 8. It can be seen from Fig. 8 that with the increase of sampling data, the estimated value of the degradation model parameter μ_0 continues to exhibit significant fluctuations. It can be seen from the estimated degradation model parameters above that, when modeling bridge performance index degradation data containing nonlinear degradation processes, the nonlinear stochastic process with a time-varying drift coefficient is superior to the linear stochastic process with a constant drift coefficient.

This study uses the method of this paper to estimate the degradation model parameter values $[\lambda_i, a_i]$ based on the degradation data from the initial time to the service time t_i, and predicts the bridge performance index values $\hat{x}_{i+1}$ corresponding to the service time t_{i+1} through $\hat{x}_{i+1} = x_i + \lambda_i\left(t_{i+1}^{a_i} - t_i^{a_i}\right)$. By connecting the predicted data points in sequence, the predicted degradation path can be obtained, as shown in Fig. 6. Among them, x_i is the actual value of the bridge performance index corresponding to the service time t_i. From Fig. 6, it can be seen that the predicted degradation paths coincide well with the simulated degradation paths, which verifies the feasibility of this paper's method to predict the value of the bridge performance index of the next sampling point. The simulated bridge performance index value corresponding to the service time $t_i = 20$ in Fig. 6 is taken as the failure threshold. With a time interval of 1, the RT and the RT probability density function to reach the failure threshold from the service time $t_i = 13$ to the service time $t_i = 19$ are predicted, as shown in Fig. 9(a), respectively. It can be seen from Fig. 9(a) that the predicted RT coincides well with the actual RT, which verifies the feasibility of this paper's method for predicting the bridge performance indexes containing nonlinear degradation processes. When the simulated degradation data is modeled by the method of using a linear stochastic process, the RT and the RT probability density function to reach the failure threshold at the same service time as in Fig. 9(a) are predicted, respectively, as shown in Fig. 9(b). It can be seen from Fig. 9 (b) that the predicted RT coincides poorly with the actual RT, and it is difficult to reliably predict the bridge performance indexes containing nonlinear degradation processes using the method of linear stochastic process, which verifies the superiority of this paper's method for predicting the bridge performance indexes containing nonlinear degradation processes.

This study takes the prediction of the RT and the RT probability density function to reach the failure threshold at service time $t_i = 16$ as an example to illustrate the prediction result calculation process after completing the degradation model parameters estimation. After completing the degradation model parameters estimation based on the degradation data from the initial moment to the service time $t_i = 16$, the estimated values of the degradation model parameters are $a = 0.5787$, $\mu_0 = 0.4096$, $\sigma_B = 0.049$, $P_0 = 0.001$, and $Q = 0$. Bringing the estimated degradation model parameter values, failure threshold, and corresponding data values at service time $t_i = 16$ into Eq. (12), the resulting Eq. (12) is the RT probability density function to reach the failure threshold. The probability values corresponding to different the RT can be calculated by Eq. (12). The RT corresponding to the maximum probability value is the predicted RT to reach the failure threshold. In addition, this study takes the prediction of the RT and the RT probability density function to reach the failure threshold at service time $t_i = 16$ as an example to analyze the impact of the initial values of the degradation model parameters on the prediction results. In this study, based on prior knowledge, a,μ_0, σ_B, P_0, and Q are simultaneously set to 0.2, 0.4, 0.6, 0.8, and 1, respectively, to predict the RT and the RT probability density function to reach the failure threshold at service time $t_i = 16$. Under the condition of assuming different initial values, the predicted RT to reach the failure threshold at service time $t_i = 16$ is 4.07, and the predicted RT probability density function to reach the failure threshold at service time $t_i = 16$ is shown in Fig. 10. It can be seen from Fig. 10 that under the condition of assuming different initial values, the prediction results remain basically unchanged, and the initial values of the degradation model parameters have little effect on the prediction results. The prediction results are not sensitive to the initial values of the degradation model parameters. In this numerical study, the average time consumption for each iteration of the algorithm is 53 s. The convergence criterion is defined as follows: using the parameter values estimated in the previous iteration as the benchmark, the algorithm is considered convergent if the mean absolute error of the currently estimated parameter values is less than 1×10^{-8}. The solution time is less affected by the data scale; under the scenario of a larger data scale, the solution time may even be shorter than that under a smaller data scale. Additionally, this study provides the recommended value ranges for the initial values of each parameter, with specific details shown in Table 2. In this paper, the degradation model parameter estimation is performed in MATLAB version 2018b. The computer is equipped with an i7–13700K CPU and 16 GB of RAM.

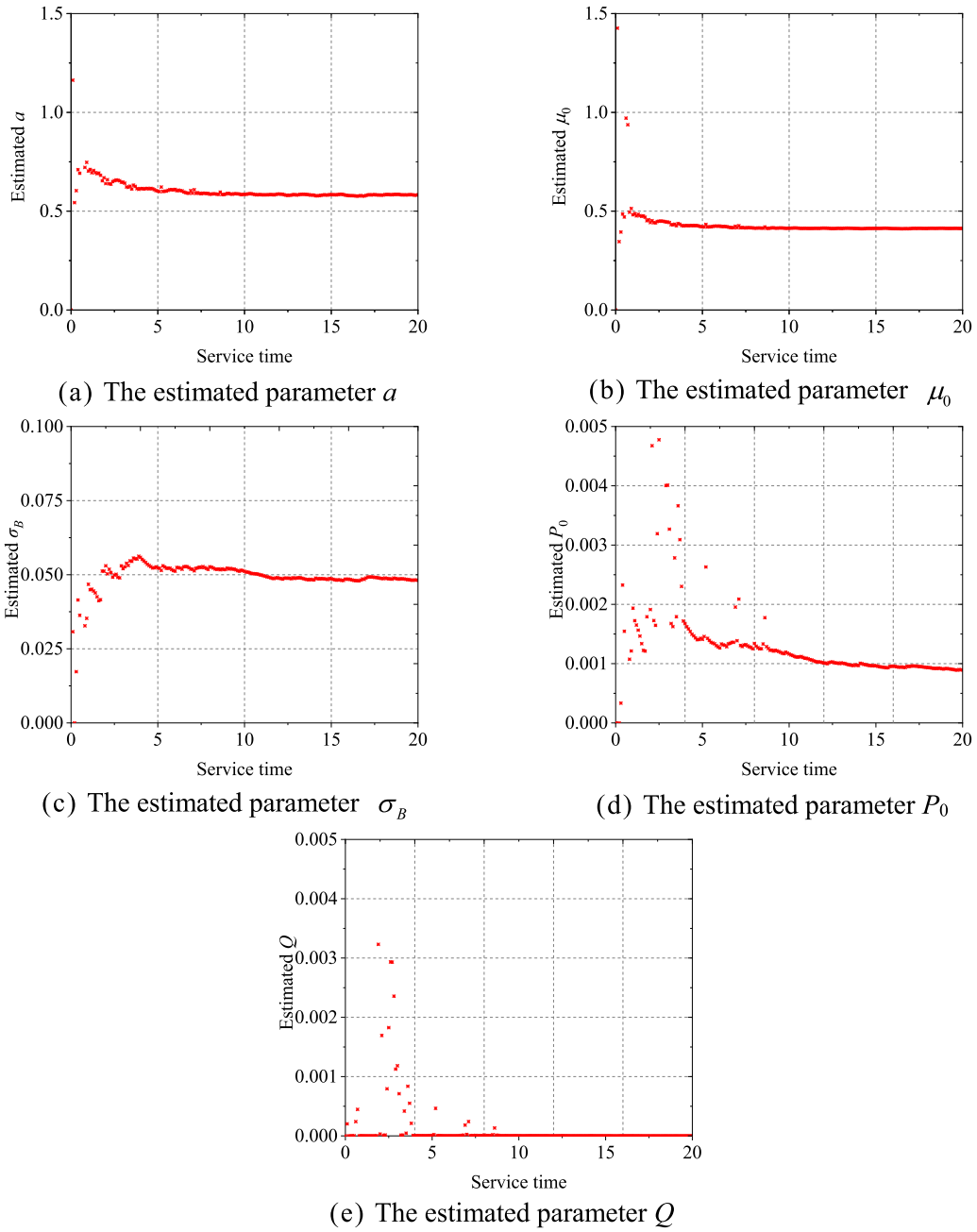


Fig. 7. Estimated values of the degradation model parameters of Eq. (2).

3.2. Study on the prediction of the bridge stiffness level

To validate the feasibility and superiority of the method of this paper in predicting the stiffness level of bridges, this study uses laboratory data to substitute for the stiffness degradation data of bridges in service. The stiffness level is one of the key performance indexes of bridges, the deflection value of the bridge under the design loads can be used as the index value to reflect the stiffness level of the bridge. When the deflection value exceeds the set limit value, it can be considered that the stiffness level of the bridge cannot meet the service requirements. Through finite element simulation of the mechanical twin model of the in-service bridge at different time periods, the deflection values under specified loads corresponding to different time periods of the in-service bridge can be readily obtained. To simulate this process, this paper conducted fatigue tests on reinforced concrete beams, simulating the long-term effects of vehicular loads on bridges through cyclically loading the reinforced concrete beams. During cyclic loading of reinforced concrete beams, a static test is conducted at fixed intervals of the number of loading cycles, and the deflection values of the reinforced concrete beams under the same static load are obtained at different numbers of loading cycles. As the number of loading cycles increases, data on the deflection growth of the reinforced concrete beam can be obtained. To conduct fatigue testing on reinforced concrete, this paper designed and constructed three reinforced concrete beams of 3.2 m in length. The test beams were constructed with concrete of strength level C50, C50 concrete is to reach a cubic compressive strength of 50 MPa on the 28th day. The tensile reinforcement is an HRB400 steel bar with a diameter of 18 mm, and the HRB400 steel bar is a hot-rolled ribbed steel bar with a yield strength of not less than 400 MPa. The compression reinforcement and stirrups are HPB300 steel bars with a diameter of 10 mm, and the HPB300 bars are hot-rolled ordinary steel bars with a yield strength of not less than 300 MPa. The test setup and specific

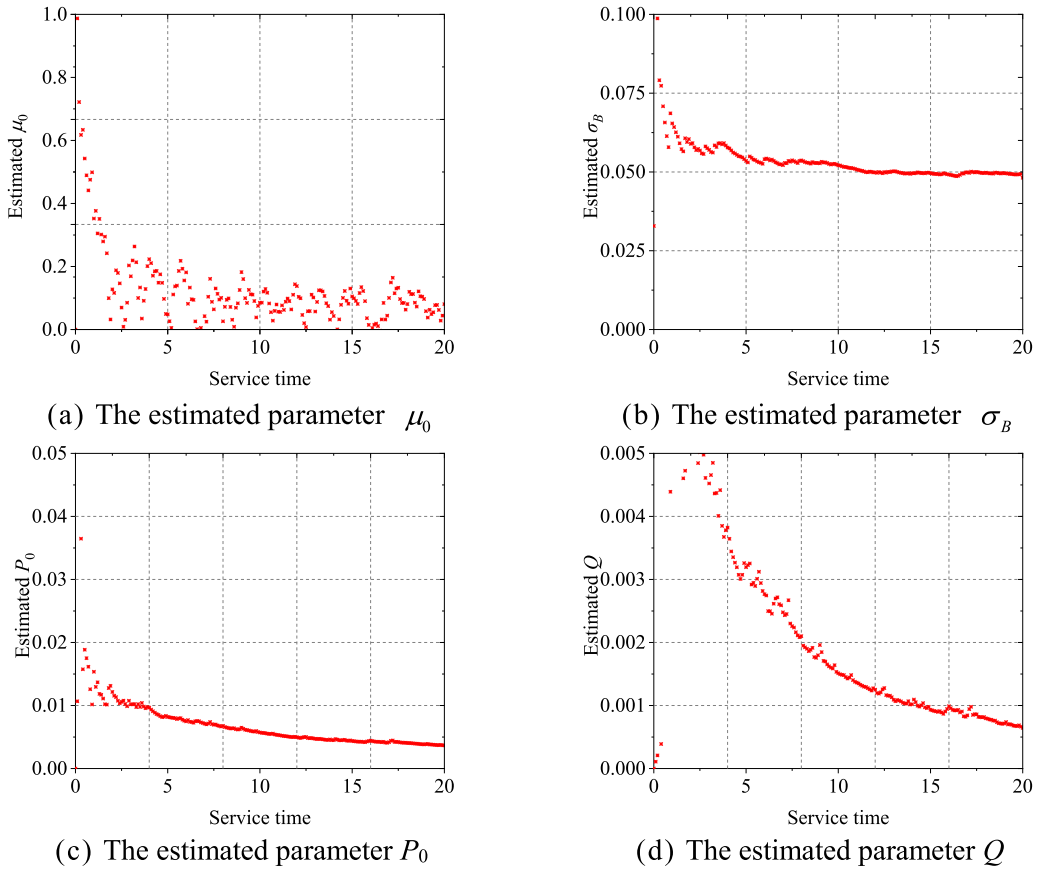


Fig. 8. Estimated values of the degradation model parameters of Eq. (1).

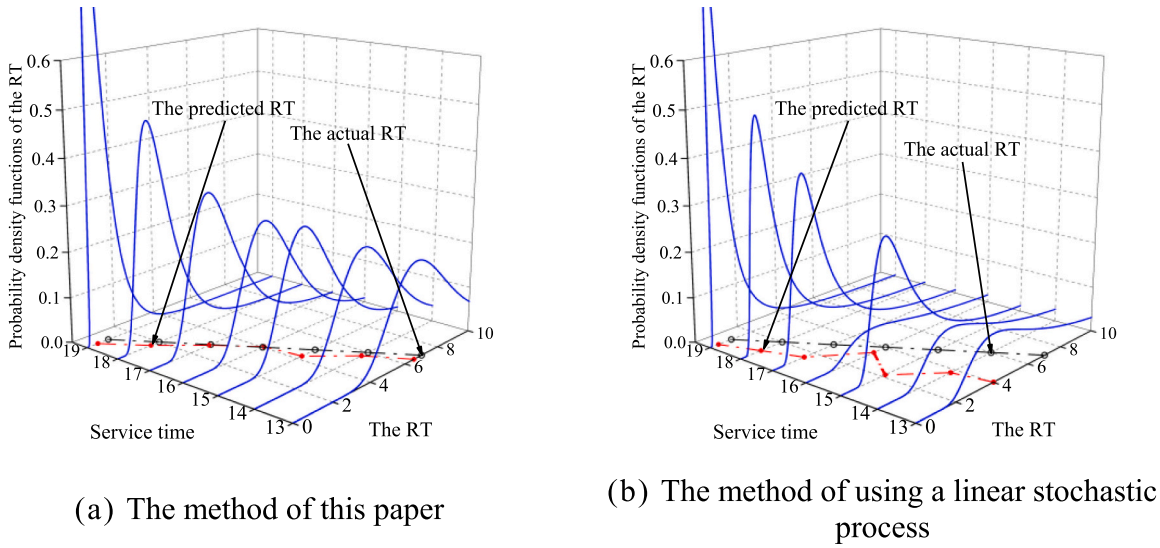


Fig. 9. The predicted RT.

parameters for the test beam are shown in Fig. 11.

In the fatigue testing of reinforced concrete beams, the three test beams were named S1, F1, and F2, respectively. The loading conditions for the three test beams are shown in Table 3, and the loading position of the test beam and the arrangement of the displacement sensors are shown in Fig. 12. Test beam S1 was loaded monotonically until failure to determine its static load capacity. Test beam F1 was loaded with a constant-amplitude cyclic load until failure, and a static test was performed on the test beam every 60,000 cycles. During each static test, the mid-span deflection value of test beam F1 at a load of 61 kN was recorded. Test beam F2 was loaded with a constant-amplitude cyclic load until failure, and a static test was performed on the test beam every 50,000 cycles. During each static test, the mid-span deflection value of test beam F2 at a load of 57 kN was recorded. Based on the recorded mid-span deflection values of test beams F1 and F2 during each static test, the change curves of mid-span deflection for test beams F1 and F2 with increasing number of cycles can be obtained. The change patterns of mid-span deflection for test beams F1 and F2 with increasing number

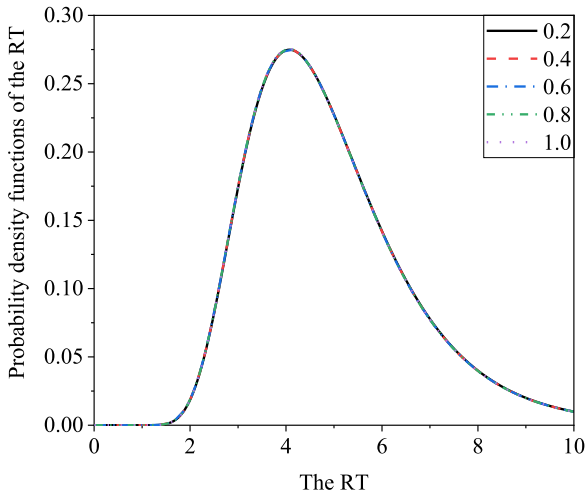


Fig. 10. The RT probability density function was calculated with different initial values.

of cycles are consistent with the test data in the literature [48]. Based on the rate of change of mid-span deflection for test beams F1 and F2 with increasing number of cycles, three stages can be divided. To validate the feasibility and superiority of the method of this paper in predicting the stiffness level of bridges, this study takes the mid-span deflection growth data from the test beams F1 and F2 in the first two stages to substitute for the degradation data of stiffness level indexes in in-service bridges. The mid-span deflection growth data for test beams F1 and F2 in the first two stages are shown in Fig. 13. Among them, test beams F1 and F2 failed after 1.57 million cycles and 2.77 million cycles, respectively.

Since $\mu(t;\vartheta)$ in the bridge performance degradation model of Eq. (2) can be any continuous function of time, the prediction results of the paper's method are obtained using the bridge performance degradation model of Eq. (29). Through the method of this paper, the mid-span deflection growth data of the test beams are modeled, the parameter values of the degradation model are estimated, and the deflection increment value corresponding to 1.5 million cycles in Fig. 13(a) is taken as the failure threshold. At an interval of 180,000 cycles, the RT and the RT probability density function to reach the failure threshold from 360,000 cycles to 1.44 million cycles are predicted, respectively, as shown in Fig. 14(a). The deflection increment value corresponding to 2.6 million cycles in Fig. 13(b) is taken as the failure threshold, and at an interval of 250,000 cycles, the RT and the RT probability density

Table 3
Loading conditions for the test beam.

Specimens	Applied load at ultimate P_u (kN)	Applied load (kN)		Applied load level P_{max} / P_u	Fatigue life N_f ($\times 10^4$ cycles)
		P_{min}	P_{max}		
S1	136.6	/	/	/	/
F1	/	13.5	61	44.7 %	157
F2	/	13.5	57	41.7 %	277

Table 2
The recommended value ranges for the initial values of each parameter.

Parameters	a	μ_0	σ_B	P_0	Q
Parameter meanings	The exponent of the power function	The coefficient of the power function	Variance	Intermediate parameters in the solution process	Intermediate parameters in the solution process
Value range	0.2–1	0.5–1.5	0.01–0.5	0–0.1	0–0.1

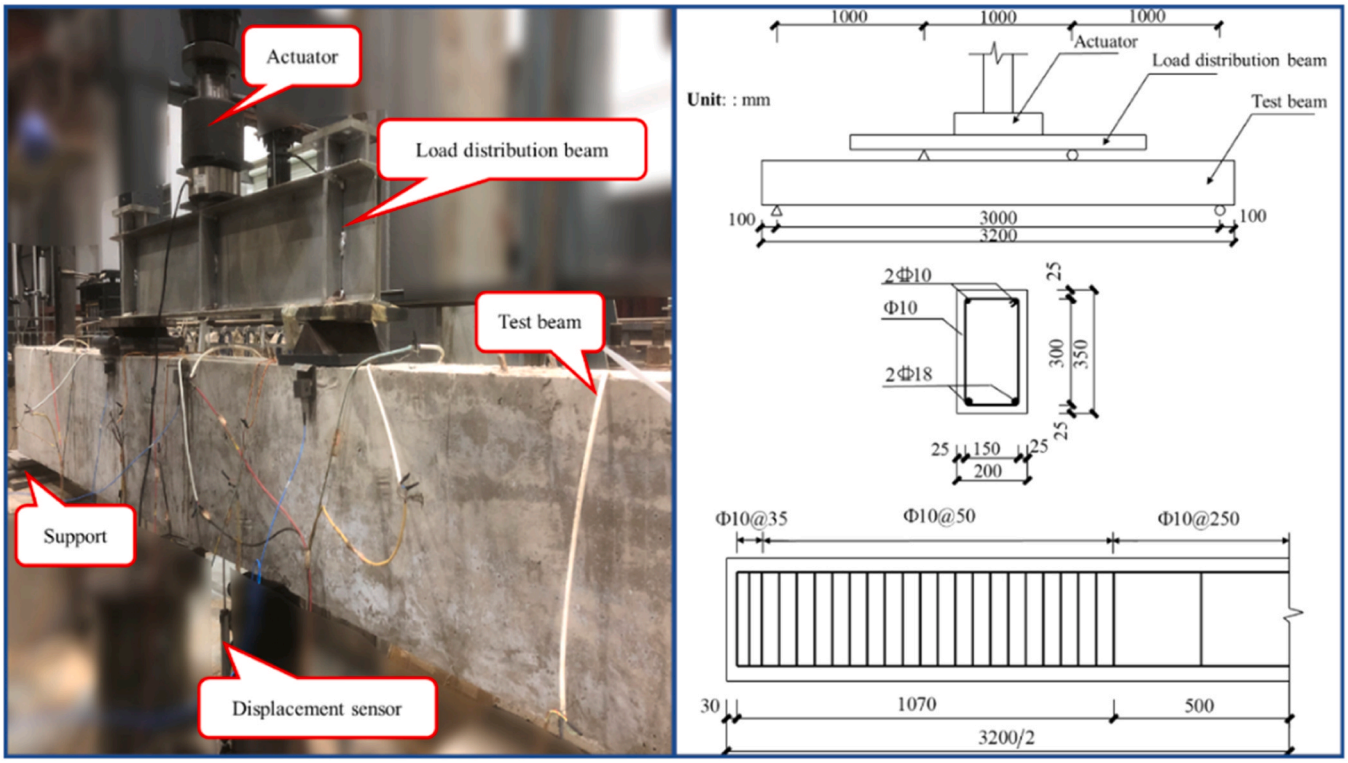


Fig. 11. The test setup and specific parameters for the test beam.

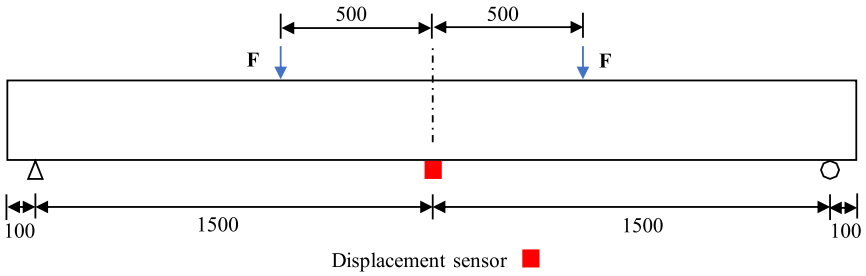


Fig. 12. Loading position of the test beam and the arrangement of displacement sensors.

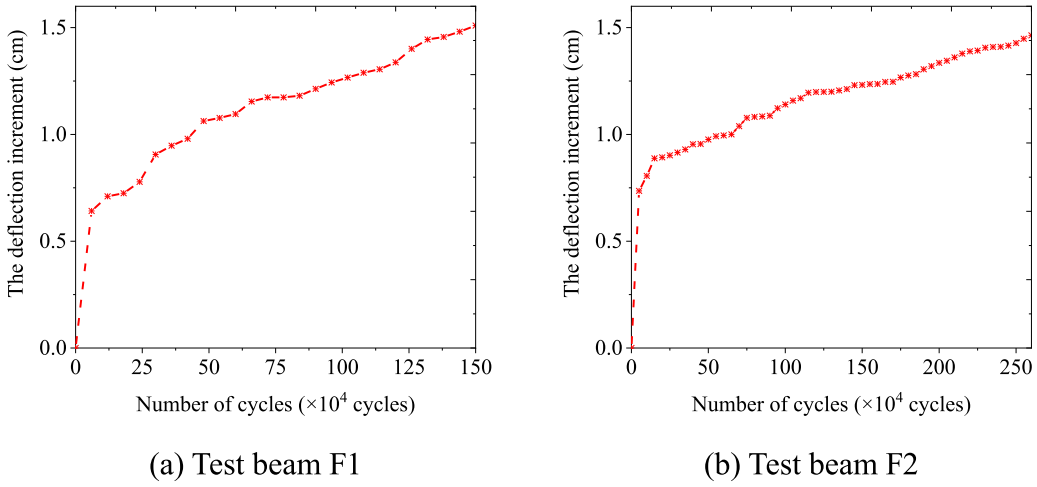


Fig. 13. Mid-span deflection growth curve of test beams.

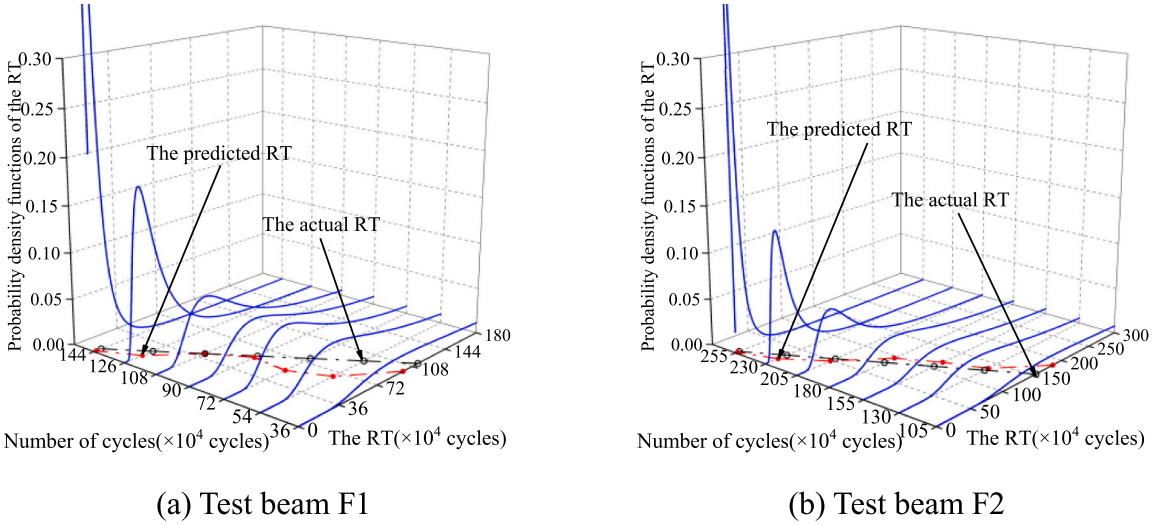


Fig. 14. Predicted RT by the method of this paper.

function to reach the failure threshold from 1.05 million cycles to 2.55 million cycles are predicted, respectively, as shown in Fig. 14(b). It can be seen from Fig. 14(a) and (b) that the predicted RT coincides well with the actual RT, which verifies the feasibility of this paper's method for predicting the bridge stiffness level.

When the mid-span deflection growth data of the test beams were modeled by the method of using a linear stochastic process, the RT and the RT probability density function at the same number of cycles as those in Fig. 14(a) and (b) were predicted, respectively, as shown in Fig. 15(a) and (b). It can be seen through Fig. 15(a) and (b) that the predicted RT coincides poorly with the actual RT, and it is difficult to realize the accurate prediction of the bridge stiffness level by the method of using a linear stochastic process, which verifies the superiority of this paper's method for predicting the bridge stiffness level.

3.3. Study on the prediction of the bridge crack severity

The severity of bridge cracks affects the durability of bridges, which leads to the shortening of bridge service life. To validate the feasibility and superiority of the method of this paper in predicting the bridge crack

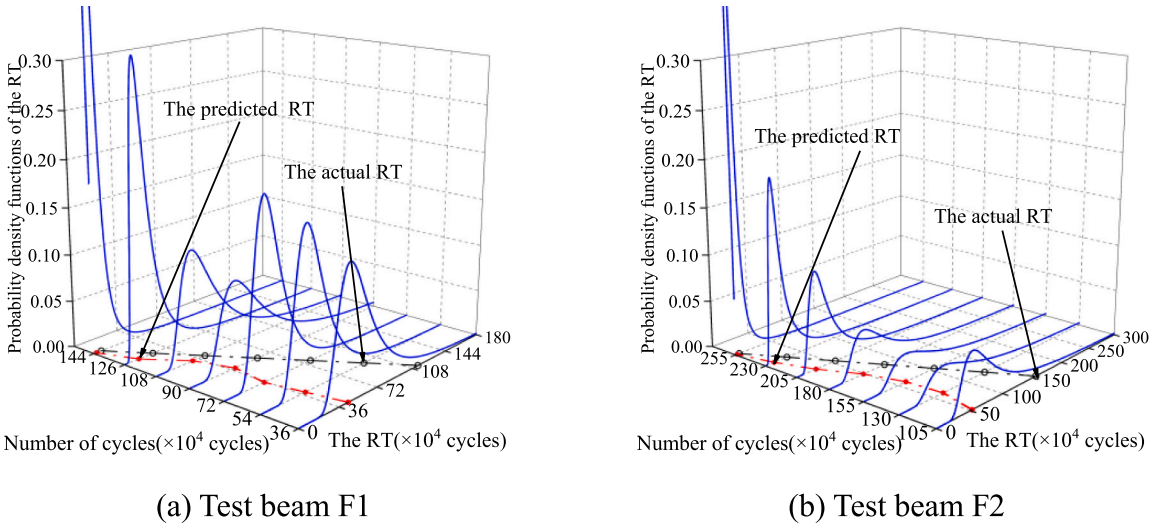


Fig. 15. Predicted RT by the method of using a linear stochastic process.

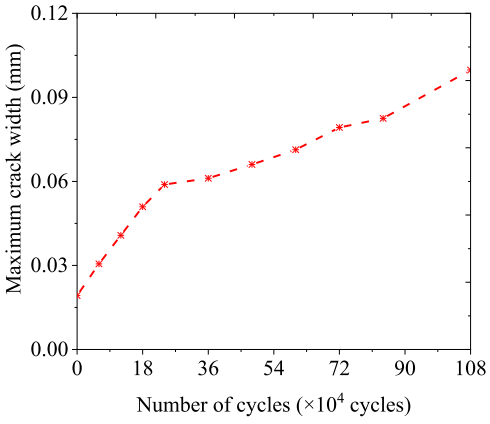


Fig. 16. Maximum crack width change curve of the test beam.

severity, this study uses laboratory data to substitute for the crack severity growth data of bridges in service. The maximum width of bridge cracks can reflect the severity of bridge cracks. When the maximum width of bridge cracks exceeds the set limit value, it can be considered that the severity of bridge cracks cannot meet the service requirements. In the literature [24], the authors carried out fatigue tests on reinforced concrete plate girders and obtained the change data of the maximum width of cracks of the test beam with the number of cyclic loading, and the crack width was measured using a handheld crack detector. The growth of the maximum width value of the crack of the test beam reflects the increase in the severity of the crack of the test beam. According to the growth rate of the maximum width value of the crack of the test beam, the development process of the crack can be divided into three stages. To validate the feasibility and superiority of the paper's method in predicting bridge crack severity, this study takes the maximum crack width growth data of the test beams in the first two stages to substitute for the degradation data of the crack severity index of in-service bridges. The maximum crack width change data of the test beam in the first two stages is shown in Fig. 16. Among them, the fatigue life of the test beam is 1.2 million cycles.

Since $\mu(t;\vartheta)$ in the bridge performance degradation model of Eq. (2) can be any continuous function of time, the prediction results of the

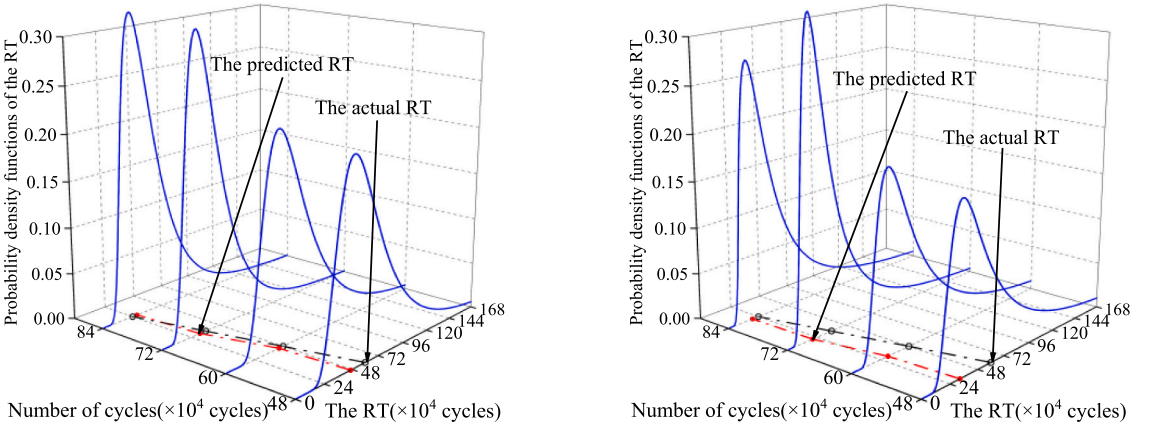
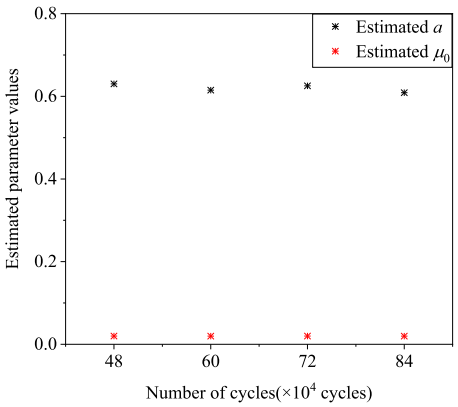
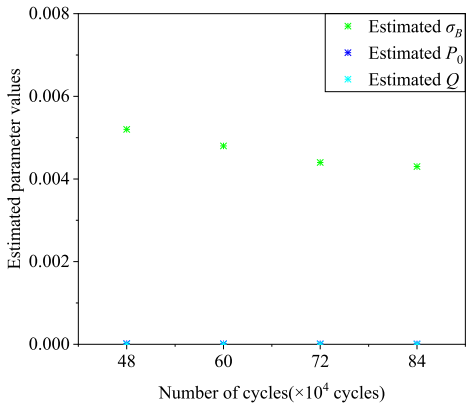


Fig. 17. The predicted RT.



(a) Estimated values of parameters a and μ_0



(b) Estimated values of parameters σ_B, P_0, and Q

Fig. 18. Estimated degradation model parameter values.

paper's method are obtained using the bridge performance degradation model of Eq. (29). Through the method of this paper, the change data of the maximum crack width of the test beam is modeled, and the parameter values of the degradation model are estimated, and the maximum crack width corresponding to 1.08 million cycles in Fig. 16 is taken as the failure threshold. At an interval of 120,000 cycles, the RT and the RT probability density function to reach the failure threshold from 480,000 cycles to 840,000 cycles are predicted, as shown in Fig. 17 (a), respectively. It can be seen from Fig. 17(a) that the predicted RT coincides well with the actual RT, which verifies the feasibility of this paper's method for predicting the bridge crack severity. When the change data of the maximum crack width of the test beam is modeled by the method of using a linear stochastic process, the RT and the RT probability density function at the same number of cycles as in Fig. 17(a) are predicted, respectively, as shown in Fig. 17(b). It can be seen through Fig. 17(b) that the predicted RT coincides poorly with the actual RT, and it is difficult to realize the accurate prediction of the bridge crack severity by the method of using linear stochastic processes, which verifies the superiority of the paper's method for predicting the bridge crack severity. In this study, the estimated degradation model parameter values obtained using the method of this paper are shown in Fig. 18. Among these, due to the significant order-of-magnitude discrepancy between the maximum crack width value and the number of cycles, the unit time adopted in modeling is the ratio of the number of cycles to 1.2 million cycles.

4. Conclusions

To fully utilize the bridge monitoring data and consider the uncertainty in the bridge performance degradation process, as well as to overcome the limitations of using the Gamma process to predict bridge performance, this paper proposes a bridge performance prediction method based on nonlinear stochastic processes. In this paper, the method was investigated using the simulated nonlinear degradation data of bridge performance indexes, the degradation data of the stiffness level of reinforced concrete beams, and the degradation data of crack severity of reinforced concrete beams, respectively. In addition, the method of using linear stochastic processes was compared with the method, and the conclusions are as follows:

1) Through the method of this paper, the simulated nonlinear degradation data of bridge performance indexes are modeled, and the parameter values of the degradation model are estimated. With the increase in sampling data, the estimated degradation model parameter values are gradually closer to the actual model parameter values, which verifies the feasibility of this paper's method for modeling the degradation data of bridge performance. Based on the estimated degradation model parameter values, this paper predicts the value of the bridge performance index at the next sampling point. The predicted degradation path coincides well with the actual degradation path, which verifies the feasibility of this paper's method to predict the value of the bridge performance index at the next sampling point. The RT and the RT probability density function of the bridge performance index to reach the failure threshold are predicted based on the estimated degradation model parameter values. The predicted RT coincides well with the actual RT, which verifies the feasibility of this paper's method for predicting the bridge performance indexes containing nonlinear degradation processes. Under the same conditions, the RT predicted by the method of using linear stochastic processes coincides poorly with the actual RT, which verifies the superiority of the paper's method for predicting bridge performance indexes containing nonlinear degradation processes.

2) Through the method of this paper, the degradation data of the stiffness level of reinforced concrete beams are modeled, and the parameter values of the degradation model are estimated. The RT and the RT probability density function of the stiffness level to reach the failure threshold are predicted based on the estimated degradation model parameter values. The predicted RT coincides well with the actual RT, which verifies the feasibility of this paper's method for predicting the bridge stiffness level. Under the same conditions, the RT predicted by the method of using linear stochastic processes coincides poorly with the actual RT, which verifies the superiority of the paper's method for predicting the bridge stiffness level.

3) Through the method of this paper, the degradation data of crack severity of reinforced concrete beams are modeled, and the parameter values of the degradation model are estimated. The RT and the RT probability density function of the crack severity to reach the failure threshold are predicted based on the estimated degradation model parameter values. The predicted RT coincides well with the actual RT, which verifies the feasibility of this paper's method for predicting the bridge crack severity. Under the same conditions, the RT predicted by the method of using linear stochastic processes coincides poorly with the actual RT, which verifies the superiority of the paper's method for predicting bridge crack severity.

4) Since the method of this paper is based on a continuous BM to predict bridge performance, it is difficult to capture the extent of bridge performance degradation caused by sudden events such as earthquakes.

When predicting multiple performance indexes of bridges, since the method of this paper does not support multi-dimensional degradation modeling, it is necessary to establish corresponding degradation models separately to achieve the prediction of each bridge performance index. Additionally, because the method of this paper requires obtaining degradation data for the bridge performance index to establish its degradation model, the method of this paper is not applicable when degradation data for the bridge performance index cannot be obtained.

CRediT authorship contribution statement

Chenxing Cui: Writing – review & editing, Supervision, Funding acquisition. **Zhiwu Yu:** Resources, Funding acquisition. **Hongshuo Sun:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Li Song:** Writing – review & editing, Supervision, Resources, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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